

A uniform Hankel obstruction for higher-order Stirling subset polynomials

A proof of the higher-order assertion in Deb–Sokal Conjecture 1.4(d)

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Abstract

Let $S_{n,k}^{(r)}$ count partitions of $n + (r-1)k$ labels into k blocks, each of size at least r , and put $s_{r,n}(x) = \sum_k S_{n,k}^{(r)} x^k$. For every integer $r \geq 3$ we prove that this polynomial sequence is not coefficientwise Hankel-totally positive. More strongly, every adjacent 3×3 Hankel determinant with positive shift has a negative coefficient of x^5 , and all its lower coefficients vanish. For shift one, the coefficient is

$$-\binom{2r-1}{r-1}^2 \frac{(r-2)(r-1)(r^2+11r+6)}{(r+1)(r+2)^2(r+3)}.$$

The proof reduces a rank-one perturbation determinant to the second differences of the two-block counting sequence. Those differences are a geometric sequence minus a partial binomial sum; an elementary ratio argument determines the sign uniformly. This proves the previously conjectured negative range of Deb–Sokal Conjecture 1.4(d), using the shift-one witness already suggested by its authors. We also obtain exact formulas for arbitrary shifts, quantitative negative bounds, small-parameter obstructions to both Stieltjes and Hamburger moment representations, and asymptotic formulas. The accompanying code supplies independent exact-arithmetic checks, not a replacement for the proofs.

Status. This is an AI-assisted research draft. The negative-range proofs are self-contained; the known positive cases $r = 1, 2$ are imported when stating the complete dichotomy. No independent referee or proof-assistant check is claimed. The source audit did not locate an intervening proof, but it does not establish priority.

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1 The problem and its relation to the supplied manifest

Deb and Sokal's Conjecture 1.4(d) asks for a sharp cutoff in coefficientwise Hankel positivity of higher-order Stirling subset polynomials: positivity at orders one and two, and failure at every higher order [1, Conjecture 1.4(d), p. 10]. Their Appendix C suggests the shift-one 3×3 determinant as a witness, without an all-order sign formula [1, p. 50]. The present argument proves that prediction and extends it to every positive shift.

The supplied manifest contains an entry entitled *A sharp order-five threshold for higher-order Stirling subset log-concavity* [3]. That entry concerns part (c) of the same conjecture: log-concavity along individual rows of the counting triangle. Here we study part (d): determinants built across different row-generating polynomials. The underlying numbers are the same, but the assertions are not duplicates. No theorem claimed by that manifest entry is used below. The manifest is treated as a catalogue of reported results, not as an independent verification of their proofs.

The useful feature of the selected problem is that a uniform negative answer can be certified by one coefficient of one fixed-size minor. There is no need to search larger and larger matrices or to control every coefficient of the row polynomials. In fact, the coefficient of x^2 in each row already contains all the information needed to obtain the obstruction.

2 Definitions and principal results

2.1 The counting triangle

For integers $r \geq 1$ and $0 \leq k \leq n$, let $S_{n,k}^{(r)}$ be the number of set partitions of an $(n + (r - 1)k)$ -element labelled set into exactly k nonempty blocks, with every block having size at least r . The empty partition gives $S_{0,0}^{(r)} = 1$. Entries outside the triangle are zero. Define

$$s_{r,n}(x) = \sum_{k=0}^n S_{n,k}^{(r)} x^k. \quad (2.1)$$

In particular, $s_{r,0}(x) = 1$, while for every $n \geq 1$,

$$s_{r,n}(x) = x + b_n x^2 + O(x^3), \quad b_n = S_{n,2}^{(r)}. \quad (2.2)$$

Here and throughout, $O(x^j)$ can be read formally as divisibility by x^j in a power-series ring. No analytic convergence is involved in coefficient extraction. The coefficient of x is one because there is exactly one partition into a single block of the required size.

For completeness, the triangular recurrence is

$$S_{n,k}^{(r)} = \binom{n + (r - 1)k - 1}{r - 1} S_{n-1,k-1}^{(r)} + k S_{n-1,k}^{(r)}. \quad (2.3)$$

To see this, distinguish the largest label. If its block has size exactly r , choose the other $r - 1$ labels and remove that block. If its block has size greater than r , remove only the distinguished label, then record which of the k blocks receives it on reinsertion. Both cases leave precisely the label counts encoded on the right-hand side.

2.2 Hankel positivity and the witness

Definition 2.1. A polynomial sequence $(p_n(x))_{n \geq 0}$ is *coefficientwise Hankel-totally positive* when every finite minor of $(p_{i+j}(x))_{i,j \geq 0}$ has only nonnegative coefficients.

We follow the source's convention that "totally positive" allows zero minors; this is also called total nonnegativity. In particular, finding one strictly negative coefficient disproves the property.

For a positive integer a , put

$$H_{r,a}(x) = \det(s_{r,a+i+j}(x))_{0 \leq i,j \leq 2}. \quad (2.4)$$

This is a minor of the full Hankel matrix: take row indices $a, a+1, a+2$ and column indices $0, 1, 2$. Let

$$K_{r,a} = [x^5]H_{r,a}(x), \quad F_m(N) = \sum_{j=0}^m \binom{N}{j}, \quad (2.5)$$

with $F_{-1}(N) = F_{-2}(N) = 0$ and $\binom{N}{-1} = 0$.

Theorem 2.2 (Uniform shifted obstruction). *Let $r \geq 3$ and $a \geq 1$ be integers. Set $N = a + 2r - 2$ and $m = r - 3$. Then*

$$H_{r,a}(x) = K_{r,a}x^5 + O(x^6), \quad (2.6)$$

$$K_{r,a} = -2^{N-1} \left(\binom{N}{m} - \binom{N}{m-1} \right) + F_m(N)F_m(N+2) - F_m(N+1)^2. \quad (2.7)$$

Moreover,

$$K_{r,a} \leq -2^{N-1} \left(\binom{N}{m} - \binom{N}{m-1} \right) < 0. \quad (2.8)$$

Consequently every determinant in (2.4) has order of vanishing exactly five at the origin and a negative leading coefficient.

Theorem 2.3 (Factored first-shift certificate). *For every integer $r \geq 1$,*

$$K_{r,1} = - \left(\frac{2r-1}{r-1} \right)^2 \frac{(r-2)(r-1)(r^2+11r+6)}{(r+1)(r+2)^2(r+3)}. \quad (2.9)$$

Thus $K_{r,1} = 0$ for $r = 1, 2$ and $K_{r,1} < 0$ for every $r \geq 3$.

Theorem 2.3 alone proves the new negative-range assertion. Theorem 2.2 explains its mechanism and gives additional consequences. These are two compatible routes to the same sign, not independent evidence from a finite list of examples.

Corollary 2.4 (The cutoff). *For positive integer r , the sequence $(s_{r,n}(x))_{n \geq 0}$ is coefficientwise Hankel-totally positive if and only if $r \in \{1, 2\}$.*

Proof. The negative range follows from either theorem. The positive cases are known results: the ordinary Stirling case and the Ward-polynomial case, respectively. The latter is supported by the continued-fraction result of Elvey Price and Sokal [2]; the deduction of the required positivity, and the status of both positive cases, are recorded explicitly in [1, p. 10]. We do not count these positive cases as new results of this article. $\square$

3 A rank-one reduction for the leading coefficient

The determinant is initially degenerate to unusually high order: after factoring one x from each row, its constant matrix is the all-ones matrix, which has rank one. Adjacent differences isolate the first nontrivial part.

Lemma 3.1 (Low-degree Hankel reduction). *Suppose $p_j(x) = x + b_jx^2 + O(x^3)$ for the five indices $j = a, \dots, a+4$. With $\Delta b_j = b_{j+1} - b_j$,*

$$\det(p_{a+i+j}(x))_{0 \leq i,j \leq 2} = x^5 \left((\Delta^2 b_a)(\Delta^2 b_{a+2}) - (\Delta^2 b_{a+1})^2 \right) + O(x^6). \quad (3.1)$$

In particular all coefficients below degree five vanish, and coefficients of p_j of degree at least three cannot affect the displayed term.

Proof. Factor x from each row. The remaining matrix is $J + xB + O(x^2)$, where J is the all-ones matrix and $B_{ij} = b_{a+i+j}$. The matrix

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

has determinant one. Multiplication on the left by L and on the right by L^T preserves the determinant. Since $LJL^T = \text{diag}(1, 0, 0)$, the resulting matrix has the block form

$$\begin{pmatrix} 1 + O(x) & O(x) & O(x) \\ O(x) & x\Delta^2 b_a + O(x^2) & x\Delta^2 b_{a+1} + O(x^2) \\ O(x) & x\Delta^2 b_{a+1} + O(x^2) & x\Delta^2 b_{a+2} + O(x^2) \end{pmatrix}. \quad (3.2)$$

The terms of degree two in its determinant are exactly the determinant of the displayed lower-right block of linear terms. All permutations using an off-diagonal entry in the first row also use an off-diagonal entry in the first column, and hence have degree at least three. Restoring the factor x^3 proves (3.1). $\square$

The sign question is therefore equivalent to a strict log-concavity inequality for the sequence $d_n = \Delta^2 b_n$. This is log-concavity in the *row index after taking two differences*, not the log-concavity of $k \mapsto S_{n,k}^{(r)}$ considered in the manifest.

4 Two-block partitions and the all-shifts proof

4.1 The two-block column

Lemma 4.1. *For $n \geq 1$ and $N = n + 2r - 2$,*

$$b_n = S_{n,2}^{(r)} = \frac{1}{2} \sum_{j=r}^{N-r} \binom{N}{j} = 2^{N-1} - \sum_{j=0}^{r-1} \binom{N}{j}. \quad (4.1)$$

The first sum is empty when $n = 1$, giving $b_1 = 0$. For $r \geq 3$,

$$d_n := \Delta^2 b_n = 2^{N-1} - F_{r-3}(N). \quad (4.2)$$

For $r = 1, 2$, instead $d_n = 2^{N-1}$.

Proof. An unordered partition into two admissible blocks is specified twice by choosing one block, whose size must lie between r and $N - r$. The omitted lower and upper binomial tails are equal by symmetry. Since $N \geq 2r - 1$, the tails are disjoint, including at the boundary $n = 1$. Subtracting them from 2^N gives (4.1).

When n increases by one, so does N . Pascal's identity gives

$$\Delta \binom{N}{j} = \binom{N}{j-1}, \quad \Delta^2 \binom{N}{j} = \binom{N}{j-2},$$

where terms with negative lower index are zero. Also $\Delta^2 2^{N-1} = 2^{N-1}$. Applying these identities term by term gives (4.2); the finite polynomial correction disappears entirely when $r = 1, 2$. $\square$

For $r \geq 3$, the relevant N satisfies $N \geq 2(r - 3) + 5$. Binomial symmetry then gives $F_{r-3}(N) < 2^{N-1}$, since its final index lies strictly below the midpoint and omits central terms. Thus $d_n > 0$; the sequence whose strict log-concavity we use is positive.

4.2 A partial-binomial log-concavity lemma

Lemma 4.2. *For fixed integer $m \geq 0$, the positive sequence $F_m(N)$, $N \geq m$, is log-concave:*

$$F_m(N)F_m(N+2) - F_m(N+1)^2 \leq 0. \quad (4.3)$$

Equality holds for $m = 0$, and the inequality is strict for $m \geq 1$.

Proof. The case $m = 0$ is $F_0(N) = 1$. Suppose $m \geq 1$. Pascal's identity gives

$$\frac{F_m(N+1)}{F_m(N)} = 1 + \frac{F_{m-1}(N)}{F_m(N)}. \quad (4.4)$$

For each $0 \leq j < m$,

$$\frac{\binom{N}{j}}{\binom{N}{m}} = \frac{m!}{j!} \frac{1}{\prod_{h=j}^{m-1} (N-h)} \quad (4.5)$$

is strictly decreasing with integer $N \geq m$. Hence so is the positive sum

$$R_m(N) = \frac{F_{m-1}(N)}{\binom{N}{m}}.$$

The ratio appearing in (4.4) is $F_{m-1}(N)/F_m(N) = R_m(N)/(1 + R_m(N))$, a strictly increasing function of $R_m(N)$. It therefore decreases with N . Successive ratios of F_m strictly decrease, which is exactly (4.3) after multiplication by the positive denominators. $\square$

This proof deliberately uses only finite sums and positive ratios. It requires no general preservation theorem for log-concavity.

4.3 Proof of Theorem 2.2

Set $B = 2^{N-1}$ and abbreviate $F_j = F_m(N+j)$ for $j = 0, 1, 2$ in this subsection only. By Lemmas 3.1 and 4.1,

$$\begin{aligned} K_{r,a} &= (B - F_0)(4B - F_2) - (2B - F_1)^2 \\ &= -B(F_2 - 4F_1 + 4F_0) + F_0F_2 - F_1^2. \end{aligned} \quad (4.6)$$

The B^2 terms cancel. Applying Pascal twice yields

$$\begin{aligned} F_m(N+1) &= F_m(N) + F_{m-1}(N), \\ F_m(N+2) &= F_m(N) + 2F_{m-1}(N) + F_{m-2}(N). \end{aligned}$$

Consequently

$$\begin{aligned} F_m(N+2) - 4F_m(N+1) + 4F_m(N) &= F_m(N) - 2F_{m-1}(N) + F_{m-2}(N) \\ &= \binom{N}{m} - \binom{N}{m-1}. \end{aligned} \quad (4.7)$$

This proves (2.7).

Here $N = a + 2r - 2 \geq 2r - 1 = 2m + 5$. If $m = 0$, the binomial difference in (4.7) is one. If $m \geq 1$, it is positive because

$$\frac{\binom{N}{m}}{\binom{N}{m-1}} = \frac{N - m + 1}{m} > 1.$$

Lemma 4.2 makes the remaining term of (4.6) nonpositive. This proves the strict negativity and the quantitative bound. The rank-one lemma proves the vanishing of the lower coefficients, so the order of vanishing is exactly five. $\square$

Remark 4.3 (The two lowest orders). For $r = 1, 2$, Lemma 4.1 makes d_n geometric. Its adjacent 2×2 determinants vanish, and hence $K_{r,a} = 0$ for every $a \geq 1$. This explains why the degree-five obstruction starts at $r = 3$. Vanishing at one degree is not itself a proof of total positivity at $r = 1, 2$; those cases remain the imported positive results of Corollary 2.4.

5 A short factored certificate at shift one

We now give a separate, very small algebraic proof of Theorem 2.3. It uses only the first five entries of the two-block column and is useful as a standalone certificate.

Let

$$A = \binom{2r-1}{r-1}, \quad u = \frac{2r}{r+1}, \quad v = \frac{2r(2r+1)}{(r+1)(r+2)}, \quad w = \frac{2r(2r+1)(2r+2)}{(r+1)(r+2)(r+3)}.$$

The recurrence (2.3) at $k = 2$ is

$$b_n = 2b_{n-1} + \binom{n+2r-3}{r-1} \quad (n \geq 2). \quad (5.1)$$

The successive binomial terms at $n = 2, 3, 4, 5$ are A, Au, Av, Aw . Starting from $b_1 = 0$ gives

$$(b_1, b_2, b_3, b_4, b_5) = A(0, 1, 2 + u, 4 + 2u + v, 8 + 4u + 2v + w). \quad (5.2)$$

Thus

$$(d_1, d_2, d_3) = A(u, 1 + v, 2 + u + w). \quad (5.3)$$

The rank-one identity immediately gives

$$\frac{K_{r,1}}{A^2} = u(2 + u + w) - (1 + v)^2. \quad (5.4)$$

For a completely explicit factorization, observe that

$$1 + v = \frac{5r^2 + 5r + 2}{(r+1)(r+2)}, \quad 2 + u + w = \frac{2(2r+1)(3r^2 + 7r + 6)}{(r+1)(r+2)(r+3)}.$$

On the common denominator $(r+1)^2(r+2)^2(r+3)$, the numerator in (5.4) becomes

$$\begin{aligned} & 4r(2r+1)(3r^2 + 7r + 6)(r+2) - (5r^2 + 5r + 2)^2(r+3) \\ &= -(r-2)(r-1)(r+1)(r^2 + 11r + 6). \end{aligned}$$

Canceling the factor $r+1$ proves (2.9). All denominators are positive for $r \geq 1$, and every factor in the numerator after the minus sign is positive for $r \geq 3$. No interpolation in r is being used: this is an identity of rational functions.

r	$\binom{2r-1}{r-1}$	$K_{r,1}$
1	1	0
2	3	0
3	10	-16
4	35	-385
5	126	-6 966
6	462	-114 345
7	1 716	-1 799 512
8	6 435	-27 756 729
9	24 310	-423 939 880
10	92 378	-6 445 028 304

6.1 The smallest higher order

$$\begin{aligned}s_{3,1}(x) &= x, \\ s_{3,2}(x) &= x + 10x^2, \\ s_{3,3}(x) &= x + 35x^2 + 280x^3, \\ s_{3,4}(x) &= x + 91x^2 + 2100x^3 + 15400x^4, \\ s_{3,5}(x) &= x + 210x^2 + 10395x^3 + 200200x^4 + 1401400x^5.\end{aligned}$$
$$H_{3,1}(x) = x^5(-16 + 19110x + 938700x^2 + 15309000x^3 + 79380000x^4). \quad (6.1)$$

For comparison, the two lower-order determinants are

$$\begin{aligned} H_{1,1}(x) &= 2x^6, \\ H_{2,1}(x) &= x^6(216 + 960x + 1440x^2 + 720x^3). \end{aligned} \quad (6.2)$$

In (6.1), the quartic after x^5 is strictly increasing for $x > 0$, starts negative, and tends to positive infinity. It has exactly one positive root ξ , and exact rational evaluations give

$$0.000805006270 < \xi < 0.000805006271. \quad (6.3)$$

$$H_{3,1}(1/2000) = -\frac{4966725131}{2560000000000000000000000} < 0. \quad (6.4)$$

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6.2 Failure of scalar moment representations

A negative polynomial coefficient does not, by itself, force the polynomial to be negative at a positive argument. Here it does force negativity *near zero*, because the negative coefficient is the first nonzero one. This distinction is essential.

Proposition 6.1 (An effective negative interval). *For $r \geq 3$ and $a \geq 1$, define $L_j = s_{r,j}(1)$ and*

$$\begin{aligned} M_{r,a} &= L_a L_{a+2} L_{a+4} + L_a L_{a+3}^2 + L_{a+1}^2 L_{a+4} \\ &\quad + 2L_{a+1} L_{a+2} L_{a+3} + L_{a+2}^3, \\ \varepsilon_{r,a} &= \min \left(1, \frac{-K_{r,a}}{2M_{r,a}} \right). \end{aligned} \tag{6.5}$$

Then $\varepsilon_{r,a}$ is a positive rational number and

$$H_{r,a}(x) \leq \frac{1}{2} K_{r,a} x^5 < 0 \quad (0 < x \leq \varepsilon_{r,a}). \tag{6.6}$$

Proof. The five products defining $M_{r,a}$ are the unsigned terms in the symmetric 3×3 determinant expansion, with the paired terms counted twice. Since each row polynomial has nonnegative coefficients, the sum of the absolute values of all coefficients of $H_{r,a}$ is at most $M_{r,a}$. For $0 < x \leq 1$ the terms of degree at least six consequently have absolute sum at most $M_{r,a} x^6$. Use $K_{r,a} < 0$ and the definition of $\varepsilon_{r,a}$. $\square$

Corollary 6.2 (Stieltjes and Hamburger exclusions). *For every $r \geq 3$, the following assertions hold. For $0 < x \leq \varepsilon_{r,1}$, the numerical sequence $(s_{r,n}(x))_{n \geq 0}$ is not a Stieltjes moment sequence. For $0 < x \leq \varepsilon_{r,2}$, it is not even a Hamburger moment sequence.*

Proof. A Stieltjes representation would give a positive measure μ supported on $[0, \infty)$ with $s_{r,n}(x) = \int t^n d\mu(t)$. For every real vector $v = (v_0, v_1, v_2)$,

$$\sum_{i,j=0}^2 v_i v_j s_{r,a+i+j}(x) = \int t^a (v_0 + v_1 t + v_2 t^2)^2 d\mu(t) \geq 0.$$

Thus the matrix defining $H_{r,a}$ would be positive semidefinite and its determinant nonnegative. Shift $a = 1$ contradicts Proposition 6.1.

For a Hamburger representation the positive measure may be supported anywhere on $\mathbb{R}$. Use $a = 2$, for which the integrand is again nonnegative. Equivalently, the matrix defining $H_{r,2}$ is the Gram matrix of t, t^2, t^3 and a principal submatrix of the unshifted Hankel matrix. Its negative determinant is again impossible. $\square$

This argument uses only the necessary Gram-matrix condition for a moment representation; no converse characterization theorem is needed. The exclusions concern sufficiently small positive x . They do not classify the numerical sequence at every positive specialization.

For a concrete Hamburger exclusion at $r = 3$, the next shifted determinant is

$$\begin{aligned} H_{3,2}(x) &= x^5 (-32 + 124614x + 12875310x^2 + 556088400x^3 \\ &\quad + 12832394400x^4 + 165904200000x^5 \\ &\quad + 1136880360000x^6 + 3202824240000x^7). \end{aligned} \tag{6.7}$$

The parenthesized polynomial has positive nonconstant coefficients, and its value at $x = 10^{-4}$ is less than -19 , as can be checked by rational arithmetic. It is therefore negative throughout $0 < x \leq 10^{-4}$. Hence the third-order sequence has no Hamburger moment representation throughout that explicit interval.

[Added 5 October 2026, batch 97G: the same Gram-matrix obstruction occurs for an Apéry-like integer sequence. Section 17 of the collection report `a279619-level-seven-gamma-constant` (in `generating-functions-and-asymptotics/oeis-sequence-asymptotics/`) proves that the Hankel matrix $(a_{i+j})_{i,j \geq 0}$ of OEIS A279619 is strictly totally positive of order five but not totally nonnegative of order six (its Theorem 17.1), and displays the failure by one polynomial witness: for an explicit $\mathcal{W}$ of degree five, the functional $x^j \mapsto a_j$ is negative on $x\mathcal{W}(x)^2$, which is nonnegative on $[0, \infty)$. It also gives the least value of a_{11} admitted by a positive measure on $[0, \infty)$ with moments $a_0, \dots, a_{10}$ (its Theorem 17.2), and proves that no tail $(a_{N+n})_{n \geq 0}$ is a Stieltjes moment sequence (its Theorem 17.17). Its positive half is certified by explicit polynomials with positive Bernstein coefficients, reduced to contiguous minors by Fekete's criterion. No theorem is shared with this report.]

7 Size, asymptotics, and useful specializations

7.1 Low-order formulas at every shift

The general formula becomes particularly simple for the first two higher orders. When $r = 3$, $m = 0$ and the partial-binomial determinant vanishes. When $r = 4$, $m = 1$ and $F_1(N) = N + 1$, whose adjacent determinant is -1 . Therefore

$$K_{3,a} = -2^{a+3}, \quad (7.1)$$

$$K_{4,a} = -2^{a+5}(a + 5) - 1. \quad (7.2)$$

These show that the obstruction is not confined to an initial corner of the Hankel matrix. Moving the test arbitrarily far down the sequence preserves the sign.

7.2 Large shift at fixed order

Proposition 7.1. *For each fixed $r \geq 3$, as $a \rightarrow \infty$,*

$$K_{r,a} \sim -\frac{2^{a+2r-3}}{(r-3)!} a^{r-3}. \quad (7.3)$$

Proof. Set $m = r - 3$ and $N = a + 2r - 2$. For fixed $m \geq 1$,

$$\binom{N}{m} - \binom{N}{m-1} = \frac{N^m}{m!} + O(N^{m-1}).$$

Each $F_m(N + j)$ is a polynomial in N of degree m , so its determinant in (2.7) is at most polynomial in N . It is negligible compared with the exponential term $2^{N-1}N^m/m!$. Since $N/a \rightarrow 1$, (7.3) follows. For $m = 0$, use the exact formula (7.1). $\square$

7.3 Large order at shift one

The rational factor in (2.9) satisfies the useful identity

$$\begin{aligned} Q(r) &:= \frac{(r-2)(r-1)(r^2+11r+6)}{(r+1)(r+2)^2(r+3)} \\ &= 1 - \frac{24r(2r+1)}{(r+1)(r+2)^2(r+3)} = 1 + O(r^{-2}). \end{aligned} \quad (7.4)$$

In particular $0 < -K_{r,1} < \binom{2r-1}{r-1}^2$ for every $r \geq 3$. Since $\binom{2r-1}{r-1} = \frac{1}{2}\binom{2r}{r}$, the elementary central-binomial asymptotic, obtained from Stirling's formula, yields

$$K_{r,1} \sim -\frac{16^r}{4\pi r} \quad (r \rightarrow \infty). \quad (7.5)$$

Thus the coefficient is not merely negative by a vanishingly small margin: its absolute size grows exponentially in the order. This does not imply a uniform interval of negative *specialized* determinants, because the higher coefficients also grow.

8 Exact verification and reproducibility

The article’s proofs establish the results for all parameter values. The supplied programs audit indexing, cancellation, integrality, factorization, and examples. They use no fitted recurrence and no floating-point sign tests.

8.1 Independent counting and determinant calculations

The standard-library program `code/verify.py` constructs the triangle using (2.3). An independent routine counts partitions by the block containing label one. Writing $T_r(M, k)$ for the number of admissible partitions of M labels into k blocks, that routine uses

$$T_r(M, k) = \sum_{j=r}^{M-r(k-1)} \binom{M-1}{j-1} T_r(M-j, k-1), \quad (8.1)$$

with $T_r(0, 0) = 1$ and impossible cases zero. This conditions on the *entire distinguished block*, rather than deleting one label as in (2.3), and checks the label shift $M = n + (r-1)k$ directly.

The determinant routine expands all six signed permutation products with exact integer polynomial arithmetic. A degree-five truncation is also used, but it still retains enough coefficients to detect the cancellation of all terms below x^5 and to compare the full six-term expansion with the two-block formula. It is not a computation of the same closed formula under a different name.

Table 2: Executed verification coverage. Counts are finite audit cases.

Check	Range	Cases
Independent partition counts	$1 \leq r \leq 6, 0 \leq n \leq 8, 0 \leq k \leq n$	270
Full polynomial determinants	$1 \leq r \leq 12, 1 \leq a \leq 8$	96
Degree-five determinant expansions	$3 \leq r \leq 30, 1 \leq a \leq 40$	1120
First-shift factorization	$1 \leq r \leq 100$	100
Partial-binomial log-concavity	$0 \leq m \leq 30, \max(1, m) \leq N \leq 100$	2665

All checks in Table 2 passed. The output records every full determinant, a grid of shifted coefficients, the first one hundred first-shift coefficients, the exact scalar examples, and a rational bracket for the root in (6.3). The optional script `code/symbolic_identity.py` independently verifies the rational factorization (5.4) using SymPy 1.14.0. That is a symbolic algebra check, not a proof-assistant certification.

8.2 Commands and file layout

The core verifier requires Python 3.9 or later and no third-party packages. From the extracted directory, run

```
python code/verify.py
```

The optional symbolic check additionally requires SymPy:

```
python code/symbolic_identity.py
```

The document uses ordinary pdfLaTeX packages and an embedded bibliography:

```
pdflatex -halt-on-error article.tex
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```

The archive also contains a `Makefile`, `README.md`, `STATUS.md`, a source audit, and a proof-audit note. The data files are written deterministically; numerical timings and machine-dependent paths are deliberately excluded from the mathematical certificates.

9 Scope and remaining questions

The established result is the all- $r \geq 3$ failure claimed in Conjecture 1.4(d), strengthened to a leading negative coefficient in *every* positive-shift adjacent 3×3 minor. Together with the previously known positive cases this gives the full dichotomy in that part of the conjecture.

Several nearby questions are not settled here. We do not prove total positivity of the numerical triangle $S^{(r)}$ itself, total positivity of any row reversal, row log-concavity, or Hankel positivity of the higher-order *cycle* polynomials. We also do not claim that all 2×2 polynomial Hankel minors are nonnegative; hence the present 3×3 witness is not asserted to have minimum possible size.

For scalar specializations, an especially concrete continuation is to classify the set of $x > 0$ for which $(s_{r,n}(x))_{n \geq 0}$ is a Stieltjes or Hamburger moment sequence. The small- x interval is now excluded, but positivity of a single determinant past its first root would not establish the required positivity of all other minors. Another question is whether an analogue of the rank-one reduction organizes leading coefficients of larger shifted minors into tractable finite-difference determinants. These are directions beyond, not conditions on, the proof above.

A Indexing and boundary audit

The most error-prone feature of this problem is that changing the number of blocks changes the number of labels in the definition of a row. For example, $S_{n,1}^{(r)}$ counts partitions of $n + r - 1$ labels, whereas $S_{n,2}^{(r)}$ counts partitions of $n + 2r - 2$ labels. Treating both as partitions of the same set would give the wrong two-block column.

The shift a in (2.4) is the index of the top-left entry, not the size of the determinant. The first witness uses polynomials $s_{r,1}$ through $s_{r,5}$. The positive-shift hypothesis is substantive: $s_{r,0} = 1$ is not divisible by x , so the rank-one argument in the form used here does not apply to shift zero.

For the all-shifts sign formula, the auxiliary parameters satisfy $m = r - 3 \geq 0$ and $N = a + 2r - 2 \geq 2m + 5$. Thus every binomial coefficient in a denominator ratio is positive when it is used. The conventions $F_{-1} = F_{-2} = 0$ and $\binom{N}{-1} = 0$ handle $r = 3, 4$ without silently invoking undefined lower indices. The cases $r = 1, 2$ are treated directly by the finite-difference identity, not by extending m to negative values in an unproved formula.

Finally, (2.9) has rational-looking factors but its value is an integer: it equals a coefficient of an integer polynomial by the proof. No denominator cancellation hypothesis or generic-parameter specialization is needed. In particular, zeros at $r = 1, 2$ are harmless, and there is no division by a Hankel determinant anywhere in the argument.

B Proof dependency map

The negative-range proof has a short dependency chain. The counting definition gives (2.2) and the two-block sum (4.1). The rank-one lemma converts the desired coefficient into $d_a d_{a+2} - d_{a+1}^2$. Pascal's identity computes d_a and the geometric-term cancellation. A ratio comparison of finite binomial sums proves the sign. Every step is an explicit algebraic or counting identity.

The first-shift route replaces the binomial-sum sign lemma with a rational factorization of three second differences. The moment corollaries add only the elementary positivity of an integral of a square and an explicit coefficient-norm estimate. The asymptotic statements add standard estimates for fixed-degree polynomials and the central binomial coefficient. Only the positive half of the final cutoff imports a nontrivial published positivity theorem.

References

- [1] Bishal Deb and Alan D. Sokal, *Higher-order Stirling cycle and subset triangles: Total positivity, continued fractions and real-rootedness*, arXiv:2507.18959v1, 2025. Conjecture 1.4(d) and discussion: p. 10; proposed 3×3 witness: Appendix C, p. 50. <https://arxiv.org/abs/2507.18959v1>.
- [2] Andrew Elvey Price and Alan D. Sokal, *Phylogenetic trees, augmented perfect matchings, and a Thron-type continued fraction (T-fraction) for the Ward polynomials*, Electronic Journal of Combinatorics **27**(4) (2020), P4.6. <https://doi.org/10.37236/9571>; <https://arxiv.org/abs/2001.01468>.
- [3] Vladimir Reshetnikov, *A manifest of docs/reports*, supplied file `manifest.tex`, 20 September 2026. Entry *A sharp order-five threshold for higher-order Stirling subset log-concavity*, in the section on log-concavity, unimodality and positivity. Used to identify a related, nonduplicate problem; not used as proof evidence.